

Assignment 1

FinTech 545 - Quantitative Risk Management

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For the moment calculations, I use denominator n and report excess kurtosis. For model comparisons, I use $AIC_c = AIC + (2k^2 + 2k)/(n - k - 1)$.

Problem 1: Reading the Shape of a Sample

Prediction

The first four sample moments are:

Moment	Value
Mean	0.001104
Variance	0.00009615
Skewness	-0.6688
Excess kurtosis	2.3398

The negative skewness means the sample has a longer left tail. The positive excess kurtosis means it has heavier tails than a Normal distribution.

- **Normal:** ruled out because its skewness and excess kurtosis are both 0.
- **Lognormal:** ruled out because its skewness should be positive.
- **Student's t :** ruled out because it is symmetric and has zero skewness.
- **NIG:** plausible because it can have negative skewness and positive excess kurtosis.

Fit and reconcile

The fitted Normal has mean 0.001104 and standard deviation 0.009806. Its 1% quantile is **-0.021707**. There are **26 observations** below this number. A correctly fitted 1% quantile should have about $1000(0.01) = 10$ observations below it.

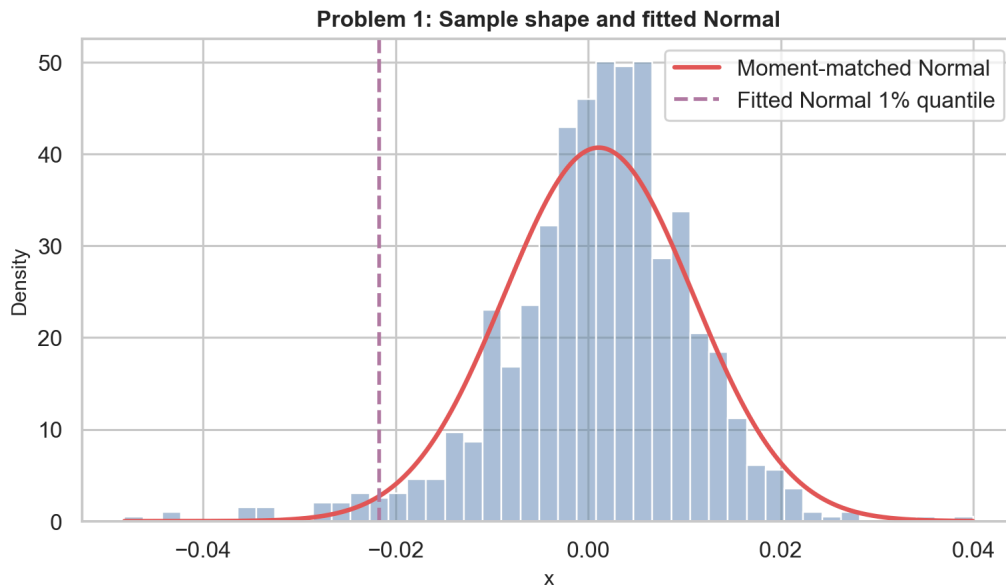


Figure 1: Sample distribution and the fitted Normal.

The result agrees with my prediction. The sample has 2.6 times as many lower tail observations as the Normal model expects. The Normal model misses the negative skewness and heavy tails, so it underestimates large negative values.

Problem 2: Regression with Non-Normal Errors

Prediction

The scatter plot shows a clear linear relationship with a few large vertical errors. Most points are close to the line, but a small number are far away. I expect a Student's t error because it allows more extreme errors than a Normal distribution.

This violates OLS assumption 7: the errors are normally distributed. If the other OLS assumptions hold, I do not expect a large or systematic change in the slope. However, the usual Normal-based tests and confidence intervals are less reliable, and large errors can make the reported standard error unstable.

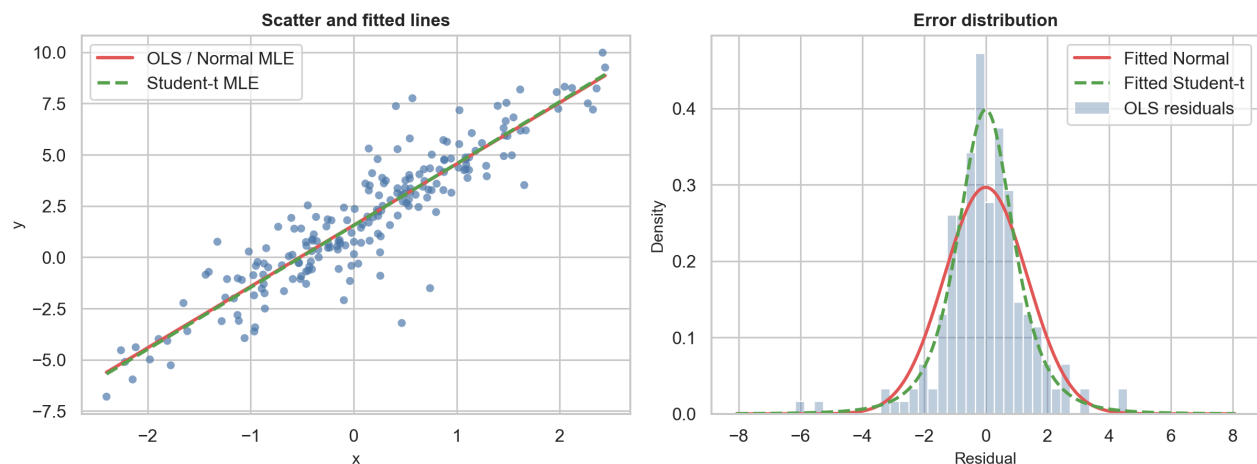


Figure 2: Scatter plot, fitted lines, and error distributions.

Fit and reconcile

Model	α	β	Error fit	AICc
OLS	1.5607	2.9908	residual SD = 1.3514	–
Normal MLE	1.5607	2.9908	$\sigma = 1.3446$	692.145
Student's t MLE	1.5480	3.0192	scale = 0.9355, $\nu = 3.6322$	665.327

The OLS standard errors are 0.0963 for α and 0.0966 for β . OLS and Normal MLE give the same coefficients because both minimize squared errors. OLS by itself does not have an AICc because it does not specify an error distribution.

The Student's t AICc is lower by 26.817, so it is strongly preferred. The slopes are still very close: 2.9908 versus 3.0192. This agrees with my prediction that non-Normal errors would not change the slope very much. The Normal model mainly gets the error shape wrong. It cannot fit both the sharp center and the extreme residuals.

Quantile	Normal error	Student's t error
95%	2.2117	2.0538
99.5%	3.4635	4.6201

At 95%, the fitted Normal is wider because the fitted t has a smaller scale. At 99.5%, the heavy tail of the t becomes more important, so the t is much wider. I would use the Student's t model for a capital buffer because capital is intended to protect against rare, extreme losses.

Problem 3: Pearson and Spearman Correlation

Prediction

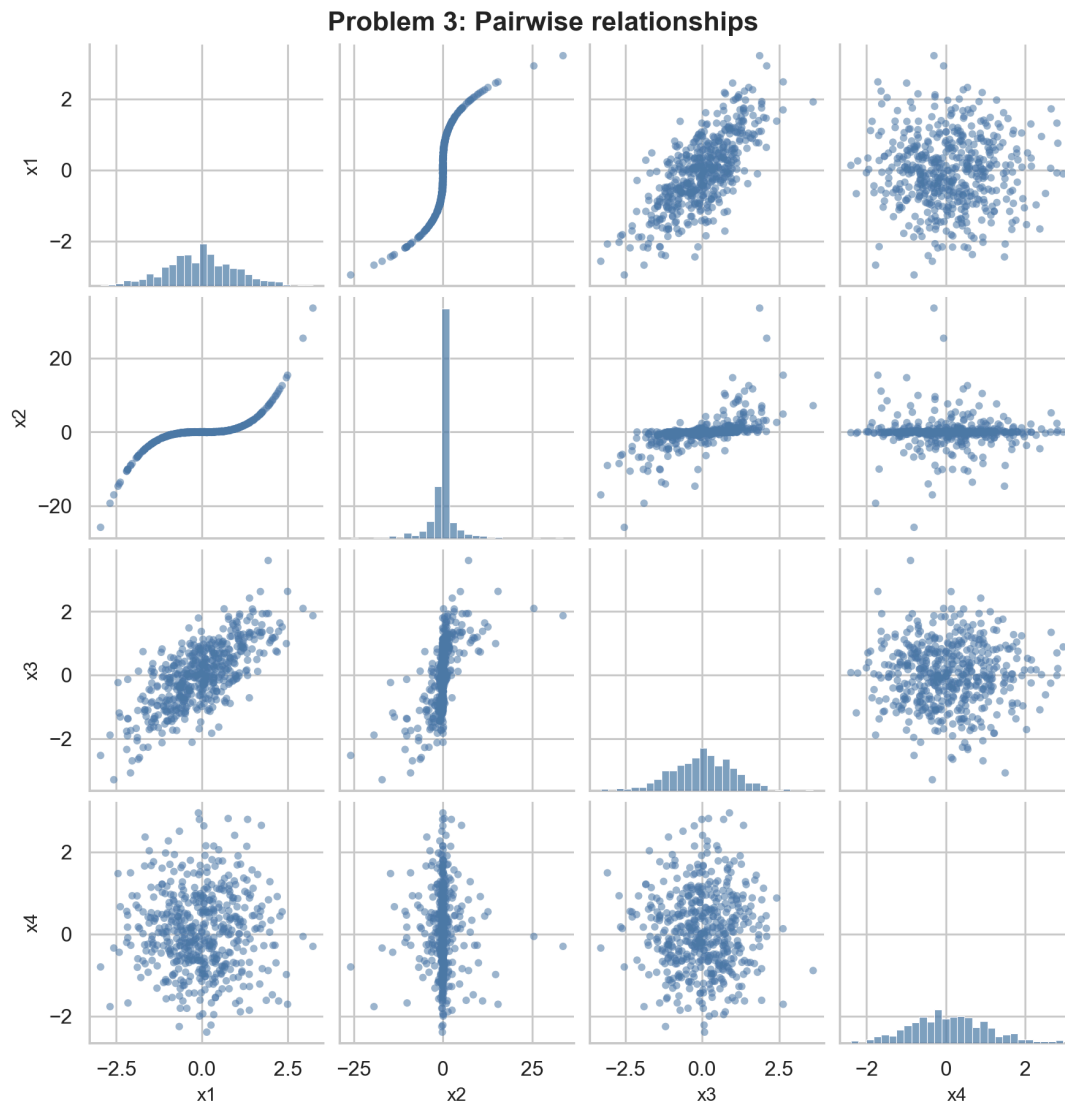


Figure 3: Pair plots for all four variables.

The x_1, x_2 plot has a strong monotonic but curved relationship. I expect Spearman to be higher than Pearson for this pair. The x_1, x_3 relationship looks approximately linear, so the two measures should be closer. Every pair with x_4 looks unrelated, so both measures should be near 0.

Fit and reconcile

Pearson correlation:

	x_1	x_2	x_3	x_4
x_1	1.000	0.799	0.719	-0.005
x_2	0.799	1.000	0.588	-0.021
x_3	0.719	0.588	1.000	-0.009
x_4	-0.005	-0.021	-0.009	1.000

Spearman correlation:

	x_1	x_2	x_3	x_4
x_1	1.000	0.972	0.684	0.000
x_2	0.972	1.000	0.667	-0.001
x_3	0.684	0.667	1.000	-0.002
x_4	0.000	-0.001	-0.002	1.000

The largest gap is for x_1, x_2 : $|0.7991 - 0.9719| = \mathbf{0.1728}$. This matches my prediction. Spearman is more honest if the question is whether the variables move in the same direction, because it measures monotonic rank dependence. Pearson answers how strong the linear relationship is. The curved shape makes Pearson lower even though the variables are strongly related.

Problem 4: Conditional Distributions

Prediction

The sample covariance matrix is

$$\hat{\Sigma} = \begin{bmatrix} 1.099875 & 1.696902 \\ 1.696902 & 4.104749 \end{bmatrix}.$$

I use $\Sigma_{11} = \text{Var}(x_1)$, $\Sigma_{22} = \text{Var}(x_2)$, and the off-diagonal blocks for the covariance. The conditional variance is

$$\text{Var}(x_2 | x_1) = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12} = 1.486746.$$

The conditional-to-unconditional variance factor is

$$\frac{\Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}}{\Sigma_{22}} = \mathbf{0.3622}.$$

Therefore, learning x_1 reduces the variance of x_2 to 36.22% of its original value. Under the multivariate Normal, this factor does not depend on the observed value of x_1 , because x_1 does not appear in the conditional variance formula.

Fit and reconcile

The conditional mean is

$$E[x_2 | x_1] = \mu_2 + \Sigma_{21}\Sigma_{11}^{-1}(x_1 - \mu_1) = 0.071073 + 1.542813x_1.$$

The coefficient 1.542813 is also the OLS slope from regressing x_2 on x_1 . The conditional standard deviation is 1.219322, which gives the constant-width 95% band shown below.

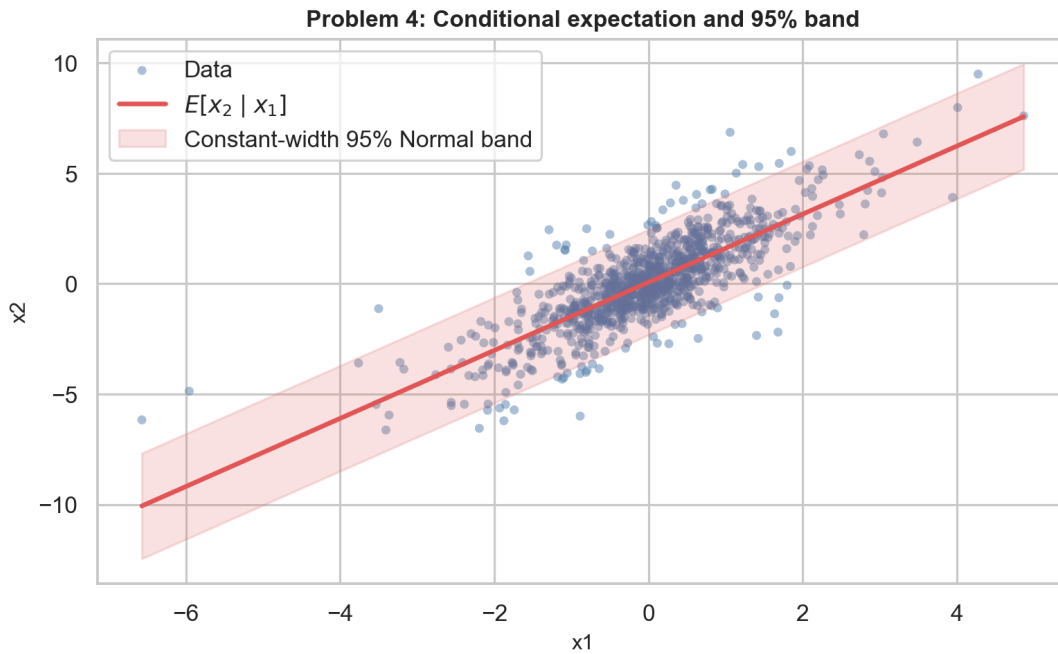


Figure 4: Conditional mean and 95% Normal band.

Overall, **935 of 1,000 observations**, or **93.5%**, fall inside the band.

Distance of x_1 from its mean	Observations	Coverage
Within 1 SD	759	95.65%
Between 1 and 2 SD	190	85.79%
Beyond 2 SD	51	90.20%

Coverage is lower away from the center. The residual standard deviation is about 1.08 in the middle and about 1.58 to 1.60 in the outer groups. This means the conditional variance is not constant, so the data do not satisfy the constant conditional variance implied by a multivariate Normal distribution.

The slope still works as the best linear regression coefficient. However, the constant band width and exact 95% Normal coverage do not remain valid. Without joint Normality, the line is also not guaranteed to be the exact conditional mean, although it is still a useful linear fit.

Problem 5: Identifying an AR or MA Order

Prediction

The series looks stationary around a constant mean. The 95% significance band is

$$\pm \frac{1.96}{\sqrt{500}} = \pm 0.08765.$$

The ACF decays with an oscillating pattern. The PACF has large spikes at lags 1 and 2 and is mostly inside the band after lag 2. The rule is:

- $AR(p)$: ACF decays and PACF cuts off after lag p .
- $MA(q)$: ACF cuts off after lag q and PACF decays.

Based on this rule, I predict an **AR(2)** process. The isolated PACF spike at lag 9 is likely sampling noise rather than a new cutoff.

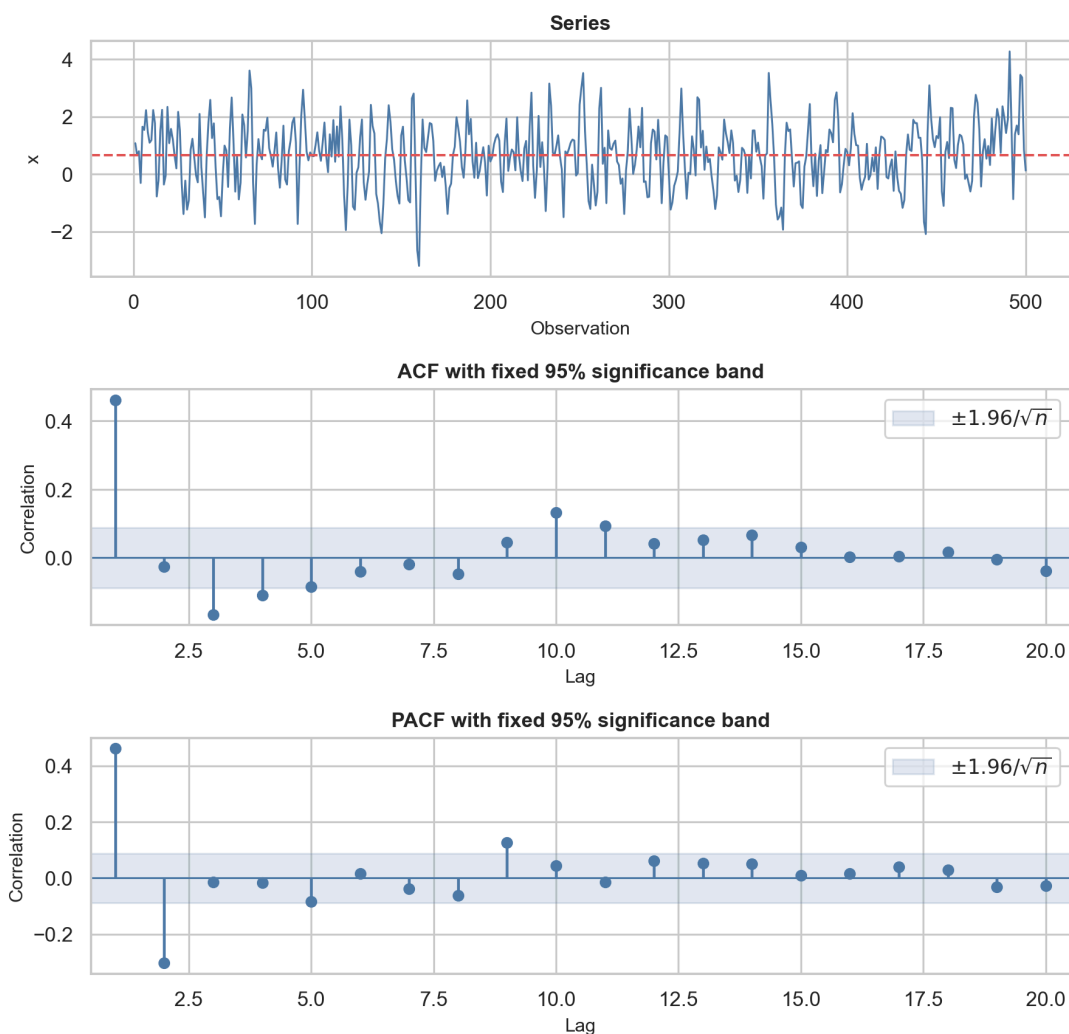


Figure 5: Series, ACF, and PACF.

Fit and reconcile

Model	AICc
AR(1)	1418.5870
AR(2)	1372.4383
AR(3)	1374.3432
MA(1)	1389.0843
MA(2)	1381.2316
MA(3)	1374.1165

AICc selects AR(2), so the fitted result matches my prediction. The AR(2) coefficients are 0.6012 and -0.3026 .

For AR(3), the third coefficient is only -0.0165 . It improves the fit very little, but AICc adds a penalty for the extra parameter. Therefore, AICc prefers AR(2). Ordinary R^2 does not include this penalty and cannot decrease when another variable or lag is added, so it would not make the same choice.